

SmartDog

Mechanical Design of a Simple Quadruped Robot

Name: Shadia Mesfer Alotaibi

1. Introduction

SmartDog is a simple quadruped robot designed using Tinkercad. The project focuses on the mechanical design of the robot including the body, legs, joints, motors, stability, and basic torque estimation.

2. Project Objective

- Design a simple four-legged robot.
- Apply basic mechanical engineering concepts.
- Understand joints, DOF, torque, and stability.

3. Mechanical Design

Body & Frame

The robot has a rectangular body made from high-strength plastic (ABS/PLA). This material is lightweight, durable, and suitable for prototype robots.

Leg Design

The robot has four identical legs. Each leg consists of a Hip Joint, Upper Leg, Knee Joint, Lower Leg, and Foot.

4. Degrees of Freedom (DOF)

Each leg has:

- Hip Joint = 1 DOF
- Knee Joint = 1 DOF

Total DOF = $2 \times 4 = 8$ DOF.

5. Motor Selection

I selected Servo Motors because they provide precise angle control, making them suitable for controlling the hip and knee joints of the robot legs.

6. Torque Estimation

Assume:

Force = 9.81 N

Distance = 0.1 m

Torque = Force $\times$ Distance

Torque = $9.81 \times 0.1 = 0.98 \text{ N}\cdot\text{m}$

7. Stability & Center of Gravity

The center of gravity is located near the middle of the robot body to improve balance and stability while standing and walking.

8. Walking Method

Walking Method: Crawl Gait

Sequence:

1. Front Right
2. Rear Left
3. Front Left
4. Rear Right

9. Expected Mechanical Challenges

- Motor overload
- Joint friction
- Leg slipping
- Balance issues
- Mechanical wear

10. Images

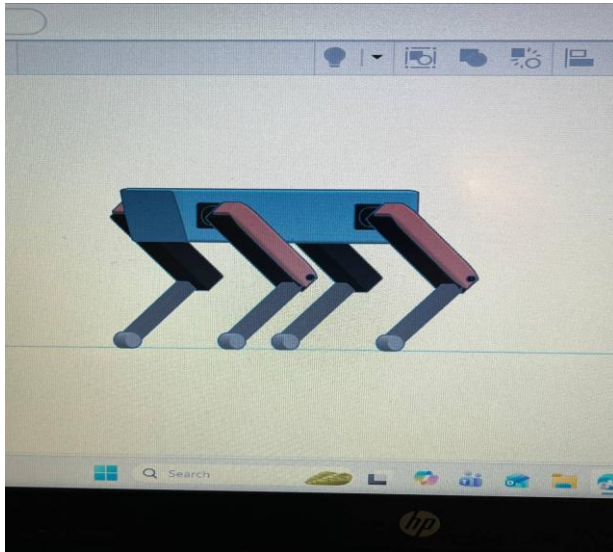


Figure 1: Isometric view of the SmartDog mechanical design.

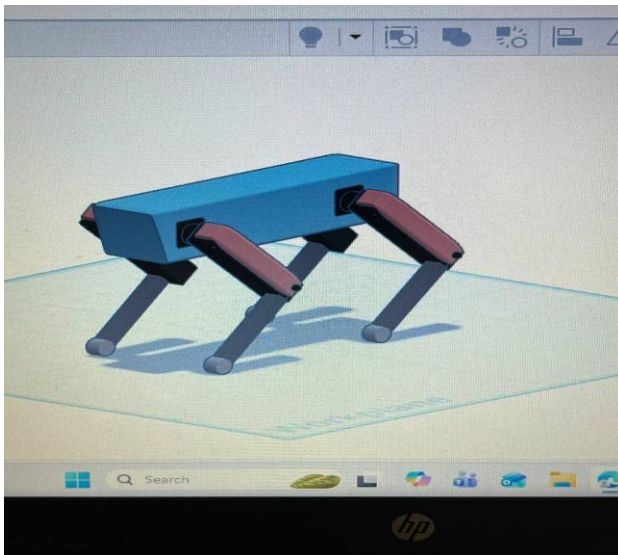


Figure 2 : Front View of the SmartDog mechanical design.

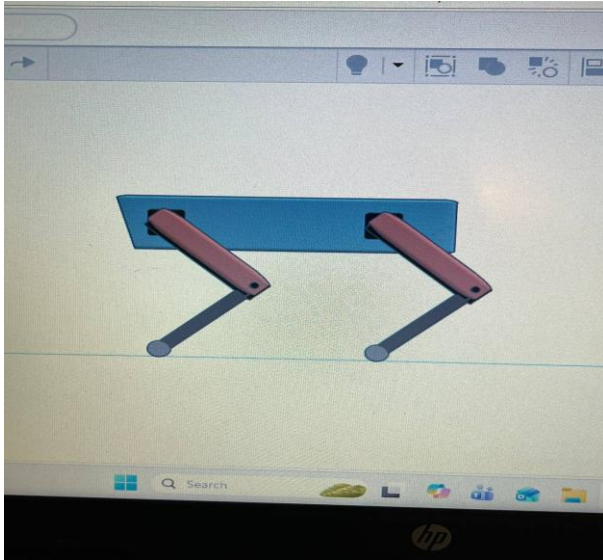


Figure 3 : Side View of the SmartDog mechanical design.

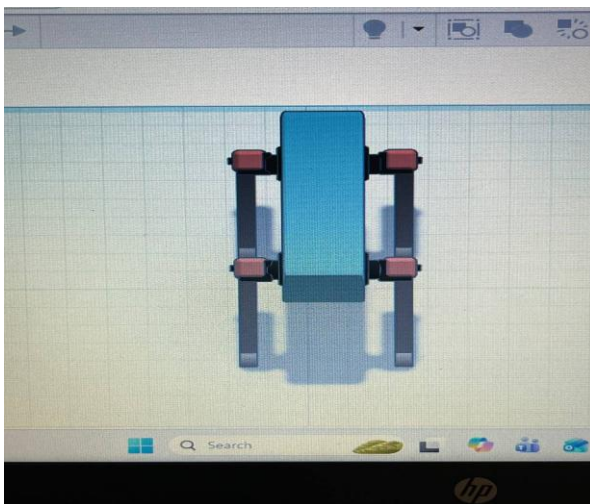


Figure 4 : Top View of the SmartDog mechanical design.

11. Conclusion

SmartDog demonstrates the basic principles of mechanical robot design using a simple, stable four-legged structure with servo motors and a lightweight body.